

Taller 4 - PostgreSQL con Docker y Docker Compose

Parte 1:

```
juanc@JHANKPC:~/cloud$ mkdir taller4
juanc@JHANKPC:~/cloud$ cd
taller4/
juanc@JHANKPC:~/cloud$ cd taller4/
juanc@JHANKPC:~/cloud/taller4$ nano docker-compose.yml
juanc@JHANKPC:~/cloud/taller4$ docker compose up -d
[+] up 2/2
✔ Network taller4_default Created 0.2s
✔ Container taller4-postgres-1 Started 1.5s
juanc@JHANKPC:~/cloud/taller4$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED       STATUS       PORTS
taller4-postgres-1  postgres:16       "docker-entrypoint.s..." postgres   21 seconds ago Up 19 seconds 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d universidad
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

universidad=#
```

Parte 2:

```
juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad
psql (16.15 (Debian 16.15-1.pgdg13+2))
Type "help" for help.

universidad=# CREATE SCHEMA academia;
CREATE SCHEMA
universidad=# CREATE TABLE academia.students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);
CREATE TABLE
universidad=# CREATE TABLE academia.courses (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    credits INTEGER NOT NULL
);
CREATE TABLE
universidad=#
```

```

CREATE SCHEMA
universidad=# CREATE TABLE academia.students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);
CREATE TABLE
universidad=# CREATE TABLE academia.courses (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    credits INTEGER NOT NULL
);
CREATE TABLE
universidad=# INSERT INTO academia.students (name, email) VALUES
('Ana', 'ana@example.com'),
('Juan', 'Juan@example.com'),
('Valentina', 'Valen@example.com');
INSERT 0 3
universidad=# INSERT INTO academia.courses (name, credits) VALUES
('Cloud Computing', 3),
('Internet of Things', 4);
INSERT 0 2
universidad=# SELECT * FROM academia.students;
 id |  name  |      email
-----+-----+-----
  1 | Ana    | ana@example.com
  2 | Juan   | Juan@example.com
  3 | Valentina | Valen@example.com
(3 rows)

universidad=# SELECT * FROM academia.students;
 id |  name  |      email
-----+-----+-----
  1 | Ana    | ana@example.com
  2 | Juan   | Juan@example.com
  3 | Valentina | Valen@example.com
(3 rows)

universidad=# █

```

Parte 3:

```

GNU nano 8.7.1 postgres/init/01-sche
CREATE SCHEMA IF NOT EXISTS academia;

CREATE TABLE IF NOT EXISTS academia.students (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.courses (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    credits INTEGER NOT NULL
);

```

```

GNU nano 8.7.1 postgres/init/02-data.sql
INSERT INTO academia.students (name, email) VALUES
('Ana', 'ana@example.com'),
('Juan', 'juan@example.com'),
('Valen', 'valen@example.com');

INSERT INTO academia.courses (name, credits) VALUES
('Cloud Computing', 3),
('Internet of Things', 4);

```

Parte 4:

```

GNU nano 8.7.1 docker-compose.yml
services:
  postgres:
    image: postgres:16
    environment:
      POSTGRES_DB: universidad
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: admin
    ports:
      - "5432:5432"
    volumes:
      - postgres_data:/var/lib/postgresql/data
      - ./postgres/init:/docker-entrypoint-initdb.d:ro

volumes:
  postgres_data:

```

Parte 5:

```

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "\dn"
List of schemas
  Name | Owner
-----+-----
academia | admin
public | pg_database_owner
(2 rows)

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "\dt academia.*"
List of relations
 Schema | Name      | Type  | Owner
-----+-----+-----+-----
academia | courses   | table | admin
academia | students  | table | admin
(2 rows)

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "SELECT * FROM academia.students;"
 id | name | email
----+-----+-----
  1 | Ana  | ana@example.com
  2 | Juan | juan@example.com
  3 | Valen | valen@example.com
(3 rows)

juanc@JHANKPC:~/cloud/taller4$

```

Parte 6:

```
GNU nano 8.7.1          postgres/init/01-schema.sql *
*CREATE SCHEMA IF NOT EXISTS academia;

CREATE TABLE IF NOT EXISTS academia.students (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.teachers (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(120) UNIQUE NOT NULL
);

CREATE TABLE IF NOT EXISTS academia.courses (
  id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  credits INTEGER NOT NULL,
  teacher_id INTEGER REFERENCES academia.teachers(id)
);

CREATE TABLE IF NOT EXISTS academia.enrollments (
  id SERIAL PRIMARY KEY,
  student_id INTEGER NOT NULL REFERENCES academia.students(id),
  course_id INTEGER NOT NULL REFERENCES academia.courses(id)
);
```

```
GNU nano 8.7.1          postgres/init/02-data.sql
INSERT INTO academia.students (name, email) VALUES
('Ana', 'ana@example.com'),
('Juan', 'juan@example.com'),
('Valen', 'valen@example.com');

INSERT INTO academia.teachers (name, email) VALUES
('Carlos', 'carlos@universidad.com'),
('Laura', 'laura@universidad.com');

INSERT INTO academia.courses (name, credits) VALUES
('Cloud Computing', 3, 1),
('Internet of Things', 4, 2);

INSERT INTO academia.enrollments (student_id, course_id) VALUES
(1, 1),
(2, 1),
(3, 2);
```

```

juanc@JHANKPC:~/cloud/taller4$ docker compose ps
NAME                IMAGE              COMMAND              SERVICE  CREAT
ED                STATUS            PORTS                ED
taller4-postgres-1  postgres:16       "docker-entrypoint.s..." postgres  31 se
conds ago         Up 4 seconds       0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "\\dt academia.*"
      List of relations
 Schema |      Name      | Type | Owner
-----+-----+-----+-----
 academia | courses        | table | admin
 academia | enrollments    | table | admin
 academia | students       | table | admin
 academia | teachers       | table | admin
(4 rows)

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "SELECT * FROM academia.teachers;"
 id | name |      email
----+-----+-----
  1 | Carlos | carlos@universidad.com
  2 | Laura | laura@universidad.com
(2 rows)

```

```

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "SELECT * FROM academia.courses;"
 id |      name      | credits | teacher_id
----+-----+-----+-----
  1 | Cloud Computing |      3 |          1
  2 | Internet of Things |      4 |          2
(2 rows)

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "SELECT * FROM academia.enrollments;"
 id | student_id | course_id
----+-----+-----
  1 |          1 |          1
  2 |          2 |          1
  3 |          3 |          2
(3 rows)

juanc@JHANKPC:~/cloud/taller4$ █

```

```

juanc@JHANKPC:~/cloud/taller4$ docker compose down
[+] down 2/2
✓Container taller4-postgres-1 Removed 0.6s
✓Network taller4_default Removed 0.4s
juanc@JHANKPC:~/cloud/taller4$ docker compose up -d
[+] up 2/2
✓Network taller4_default Created 0.1s
✓Container taller4-postgres-1 Started 1.0s
juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "SELECT * FROM academia.enrollments;"
 id | student_id | course_id
-----+-----+-----
  1 |          1 |          1
  2 |          2 |          1
  3 |          3 |          2
(3 rows)

juanc@JHANKPC:~/cloud/taller4$ docker compose exec postgres psql -U admin -d
universidad -c "
> SELECT
>     s.name AS student,
>     c.name AS course,
>     t.name AS teacher
> FROM academia.enrollments e
> JOIN academia.students s ON e.student_id = s.id
> JOIN academia.courses c ON e.course_id = c.id
> JOIN academia.teachers t ON c.teacher_id = t.id;
> "
 student |      course      | teacher
-----+-----+-----
 Ana     | Cloud Computing  | Carlos
 Juan    | Cloud Computing  | Carlos
 Valen   | Internet of Things | Laura
(3 rows)

juanc@JHANKPC:~/cloud/taller4$ █

```